Biosensors and Bioelectronics 25 (2010) 1109–1115

Contents lists available at ScienceDirect

Biosensors and Bioelectronics

journal homepage: www.elsevier.com/locate/bios

On-chip electrical impedance tomography for imaging biological cells

Tao Sun[a,*,1], Soichiro Tsuda[b,1], Klaus-Peter Zauner[b], Hywel Morgan[a,*]

[a] Nano Research Group, School of Electronics and Computer Science, University of Southampton, SO17 1BJ, United Kingdom
[b] Science and Engineering of Natural Systems Group, School of Electronics and Computer Science, University of Southampton, SO17 1BJ, United Kingdom

ARTICLE INFO

Article history:
Received 9 July 2009
Received in revised form 4 September 2009
Accepted 23 September 2009
Available online 2 October 2009

Keywords:
Electrical impedance tomography
Cell culture
Cell imaging
Lab-on-a-chip

ABSTRACT

Electrical impedance tomography is an imaging technology that spatially characterizes the electrical properties of an object. We present a miniaturized electrical impedance tomography system that can image the electrical conductivity distribution within a two-dimensional cell culture. A chip containing a circular 16-electrode array was fabricated using printed circuit board developing technology and used to inject current and to measure spatial voltage across the object. The signal stimulation and voltage data acquisition were performed using an impedance analyzer, operating in four-electrode mode. An open source software, EIDORS was used for image reconstruction. Finite element modelling was used to simulate the image reconstruction process by imaging two ellipsoidal phantoms in the circular 16-electrode array. The effect of the regularization parameter in the reconstruction algorithm and the influence from noise on the fidelity of the images has been numerically analyzed. Experimentally, we show reconstructed images of a multi-nuclear single cellular organism, *Physarum Polycephalum*, demonstrating the first step towards impedance imaging of single cells in culture. Our system provides a non-invasive lab-on-a-chip technology for spatially mapping the electrical properties of single cells, which would be significant and useful for diagnostic and clinical applications.

1. Introduction

Electrical impedance spectroscopy (Schwan, 1957; Pethig, 1979; Morgan et al., 2007; Sun et al., 2007) is a high speed, non-invasive technique that is used to characterize the dielectric properties of biological cells. In the simplest sense, a biological cell consists of a conducting cytoplasm (with nucleus) covered by an ultra thin insulating cell membrane. When exposed to an ac electric field, at low frequencies the cell behaves as an insulating object. The cell membrane is equivalent to a capacitor and prevents the electric field lines from penetrating the cell. Therefore information concerning cell size (and shape) can be obtained by measuring the electrical properties of the cell at low frequencies. At higher frequencies, the cell membrane electrically becomes shorted-circuited. Thus the internal properties of the cell (i.e. cytoplasmic resistance) can be probed. For adherent cells, the kinetics during culture, such as cell growth, differentiation, and the effect of drugs can be monitored using electrical impedance. Cells are grown on the surface of an array of electrodes and the impedance is monitored as a function of time. This technique is termed electric cell-substrate impedance sensing (ECIS) (Wegener et al., 2000; Keese et al., 2004; Blasio et al., 2004).In electrical impedance tomography (EIT) (Borcea, 2002; Holder, 2005; Bayford, 2006), measurements of a sample are made between multiple electrodes. The technique generates images of the conductivity distribution inside an object. Although, the spatial resolution of EIT cannot be compared with magnetic resonance imaging (MRI), the technique is harmless, low cost and has good temporal resolution. Theoretically, EIT is an inverse problem: given an object with unknown electrical properties, reconstruction of an image requires determination of the internal conductivity (or permittivity) profile of the object from a finite number of boundary measurements (current/voltage). Over the past 20 years, a large number of reconstruction algorithms have been developed to improve the computational efficiency, reconstruction sensitivity and image fidelity. Examples include the modified Newton–Raphson method (Yorkey et al., 1987), back-projection algorithm (Santosa and Vogelius, 1990), variational method (Kohn and Mckenney, 1990), maximum a *posteriori* approach (Adler and Guardo, 1996), Monte Carlo sampling method (Kaipio et al., 2000) and direct reconstruction algorithms (Mueller et al., 2002). Most bio-medical and clinical uses of EIT are *in vivo* and used to detect disease such as breast cancer (Cherepenin et al., 2001) or to monitor brain function (Bagshaw et al., 2003). Recent interest has centred on *in vitro* applications, e.g. Davalos et al. (2002, 2004) proposed using EIT for detecting and imaging electroporation of cells in tissue in real-time. Recently, there has been an interest in miniaturizing the technology. Hou et al. (2007) patterned 32 electrodes in a rectangular configuration to provide a two-dimensional map of the conductivity of carbon nanotube thin films. Chai et al. (2005, 2007)

* Corresponding authors. Tel.: +44 2380 593330; fax: +44 2380 593029.
E-mail addresses: ts5@ecs.soton.ac.uk (T. Sun), hm@ecs.soton.ac.uk (H. Morgan).
[1] Note that T. Sun and S. Tsuda contributed equally to this paper.

0956-5663/$ – see front matter © 2009 Elsevier B.V. All rights reserved.
doi:10.1016/j.bios.2009.09.036

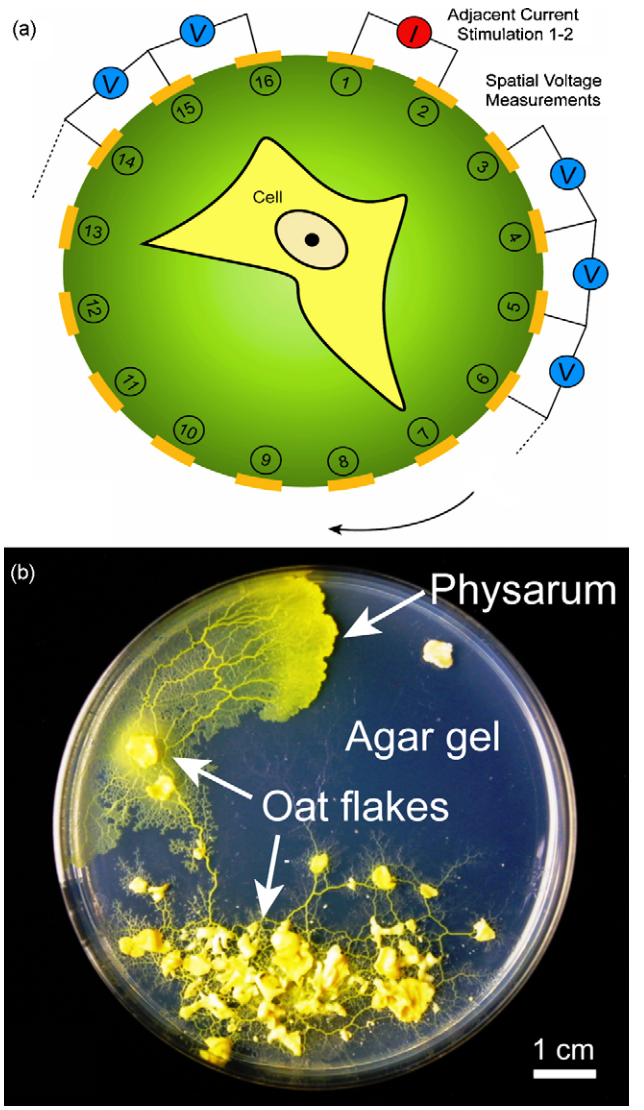


Fig. 1. (a) Diagram showing the principle of electrical impedance tomography (EIT) for imaging single cell in culture in a circular electrode pattern. (b) Photograph showing *Physarum polycephalum* cultured (fed with oat flakes) on the surface of agar gel.

fabricated a set of 4 × 4 complementary metal oxide semiconductor microelectrode array to image a cell culture environment. However, the image quality was limited by the number of electrodes (voxels). Linderholm et al. (2006, 2007, 2008) used an array of interdigitated electrodes to monitor the migration, stratification and influence of detergent on cells in culture. In these papers (Chai et al., 2007; Linderholm et al., 2007, 2008), the electrodes needed to be covered with confluent cell monolayers in order that the conductivity distribution could be mapped. Also, the direct contact between the cells and the electrodes may lead to the damage to the cells due to the local high electric field intensity and the Joule heating effect.

In this paper, we fabricated an imaging chip consisting of 16 equal-spaced electrodes in a circular pattern around a cell culture chamber (Fig. 1a). The cell can be cultured at any position within the circular array for imaging. To perform EIT, a stimulating signal is injected through one pair of electrodes and the voltage across all sequential electrode-pair combinations measured. The electrical stimulations and measurements are non-contact to the cell. The perturbation in the conductivity distribution in the system is characterized by the voltage data, and images the cell are reconstructed using software termed "Electrical Impedance Tomography and Diffuse Optical Tomography Reconstruction Software", EIDORS (Polydorides and Lionheart, 2002; Adler and Lionheart, 2006). We analyzed the fidelity of image reconstruction by modelling two ellipsoidal phantoms within the system and demonstrated the first step towards single cell impedance tomography by imaging a large multi-nuclear single-cell organism – the amoeboid plasmodium of the slime mold *Physarum Polycephalum*, Fig. 1b.

2. Electrical impedance tomography (EIT)

The mathematical theory of EIT comprises two parts: the forward problem and the inverse problem.

The forward problem corresponds to the calculation of the potentials on the electrodes for a known injected current and conductivity distribution. Solving the forward problem involves constructing the global Jacobian (sensitivity) matrix of the system, which is required for image reconstruction and the solution of the inverse problem. For EIT, the electric field distribution in considered quasi-electrostatic. In this work, only low frequency ac is used and the analysis is restricted to mapping the conductivity distribution of the object. Therefore, the electrical potential ϕ, satisfies Laplace's equation:

$$\nabla \cdot \sigma \nabla \phi = 0 \tag{1}$$

Eq. (1) gives a unique solution for the electric potential distribution, if the boundary conditions are defined. The forward problem is well-posed in terms of existence, uniqueness and stability. However, in the case of EIT, since the potential distribution is dependent on the conductivity distribution, it cannot be solved analytically for arbitrary values of σ. Therefore a general approach is to set up a numerical model that can simulate the potential values on the electrodes as close as possible to the measured voltage data in an experiment. The numerical model discretizes the domain into small elements so that an approximate solution can be obtained. Image reconstruction requires knowledge of the conductivity distribution. In EIT, the change in conductivity σ_{q} within the domain, and the change in the potential distribution ϕ_{p} are linked by the Jacobian (sensitivity) matrix $\mathbf{J}_{\mathrm{p,q}}$:

$$\mathbf{J}_{\mathrm{p,q}} = \frac{\partial \phi_{\mathrm{p}}}{\partial \sigma_{\mathrm{q}}} \tag{2}$$

For computational efficiency, EIDORS calculates the Jacobian as the dot product of the current and the measurement field multiplied by the integral of the gradients of the weighted functions over every element (Polydorides and Lionheart, 2002).

The inverse problem corresponds to calculation of the conductivity distribution when the injected current is known and the voltages are measured. The inverse problem is "ill-posed" since the solutions of the inverse problem are not unique. The process of solving the inverse problem requires finding a stable value of σ, so that the difference between the numerical simulations $\phi(\sigma)$ and the measured voltage V is minimised.

$$f(\sigma) = \tfrac{1}{2}(\phi(\sigma) - V)^*(\phi(\sigma) - V) = \tfrac{1}{2}||\phi(\sigma) - V||^2 \tag{3}$$

To minimize Eq. (3), the linear generalized Tikhonov regularization can be encapsulated into the Gauss–Newton algorithm, which gives calculation of $\Delta\sigma$ as (Polydorides and Lionheart, 2002)

$$\Delta\sigma = [\phi'(\sigma)^*\phi'(\sigma) + \lambda^2 R^* R]^{-1}(\phi'(\sigma)^* V + \lambda^2 R^* R\sigma) \tag{4}$$

where R is either the identity matrix or other regularization matrix including certain *image prior* assumptions about σ for reconstruction. λ is a positive scalar regularization parameter. The actual use of R and λ in EIDORS software will be clarified in the next section.

3. Materials and methods

3.1. *Physarum polycephalum*

The plasmodium of the slime mold *P. polycephalum* is an amoeboid multi-nucleated single cellular organism. It consists of fan-like advancing fronts connected by tubular structures found in the posterior region. This organism has been long used as a model organism for the study of mitosis (Daniel and Rusch, 1961; Dove and Rusch, 1980; Sauer, 1982), membrane potential (Sudbery and Grant, 1976; Hato, 1979) and protoplasmic streaming (Wohlfarth-Bottermann, 1979; Fingerle and Gradmann, 1982). A *Physarum* plasmodium can extend to over several centimetres in diameter; nevertheless it still maintains itself as a single cell by protoplasmic flow. Being a multi-nucleated cell, even if a cell is split into pieces, each piece can keep alive and act as single cell. It can be regarded as an autonomous distributed system and recently, its oscillatory behaviour has been utilized directly for robotic control applications (Tsuda et al., 2007).

We employed *P. polycephalum* for EIT imaging for the following reasons. First, if a small portion of *Physarum* is cut out from a larger culture, it self-repairs and behaves as a single cell, so that single *Physarum* cells can be made to an arbitrary size. Second, *Physarum* cell can grow on an agar gel (nutrient free) for several hours. This property is suitable for imaging the kinetics of cell growth and migration. *P. polycephalum* was cultured on 1.5% agar gel in a petri-dish. It was kept in the dark at room temperature ($22 \pm 2\,^{\circ}$C) and regularly fed with oat flakes (once a day). When a *Physarum* cell covered the agar gel, a section of the cell ($6\,\text{cm} \times 6\,\text{cm}$) was cut together with the agar and transferred to a new agar gel plate. Prior to experiments it was starved for 12 h. A small portion of a cell (fresh weight 15 mg) excised from an actively growing front region of a large plasmodium was used for the EIT measurement.

3.2. Chip fabrication

The EIT chip contains 16-electrodes positioned in a circular pattern, and made from a printed circuit board (PCB). The electrode length and spacing are identical, the electrodes are 35 μm thick and the area of an individual electrode is 2.06 mm^2. A thin gold layer was galvanostatically plated onto the surface of the copper electrodes using a three-electrode potential station (PGSTAT128N, Autolab electrochemical instruments, Netherlands) with a platinum-coated metal mesh counter electrode and Ag/AgCl electrode (REF321, Radiometer Analytical S.A., France) reference electrode. The thickness of the gold layers was 500 nm and controlled by adjusting the deposition time. To culture the plasmodium, a polydimethylsiloxane (PDMS) sheet (1 mm thick) with a 6 mm diameter hole was clamped over the EIT chip. A thin layer of agar gel was deposited over the surface of the electrodes, within the PDMS chamber, over which, the *P. polycephalum* was cultured. This avoids direct contact between the electrodes and the *P. polycephalum*. To eliminate evaporation, a second PDMS sheet and a polymethylmethacrylate (PMMA) plexiglass lid was used as shown in Fig. 2.

3.3. System instrumentation

Signal stimulations and measurements were performed using an Alpha-A impedance analyzer (Novocontrol Technologies, Germany) operating in four-electrode impedance measurement mode. A custom-designed electronic platform was built for multiplexed measurements, switching the pattern of stimulating signal and measuring voltages on different combinations of electrode-pairs. Analogue chips, consisting of 16×8 cross-point switches (Intersil CD22M3494) were used as multiplexers. The impedance analyzer and the multiplexing board were both controlled by a Linux PC (Fedora 7) through a GPIB-PCI interface (NI PCI-GPIB, National Instruments, USA) and a USB controller (FT2232D, FTDI Ltd. UK), respectively. Custom software (written in C) was used for automated real-time measurements. The software sets the amplitude and frequency of the stimulating signal, the digital signals to the address lines of the multiplexers and controls the data acquisition. The highest temporal resolution of individual voltage measurement is 100 ms, which is limited by the data rate of the impedance analyzer.

3.4. EIT measurement scheme

The adjacent method was used for signal stimulation and voltage measurements. This method is more sensitive to variations in the impedance of objects near the electrodes than in the centre of the chip. The stimulating signal is first applied through electrodes 1 and 2 (Fig. 1) and the voltage measured from successive electrode-pairs: 3–4, 4–5, . . ., 15–16, giving 13 voltage measurements. Note that voltages across electrode-pair 16–1, 1–2 and 2–3 cannot be measured since the stimulating signal is injected from electrode-pair 1–2. Then a second set of measurements is made with the current applied through electrodes 2 and 3 giving a further 13 sets of voltage data. This process is repeated, generating a total of $16 \times 13 = 208$ voltage measurements, which requires approximately 20 s for a full scan. Owing to reciprocity, only half of the measurements are linearly independent, all the data are used in the image reconstruction.

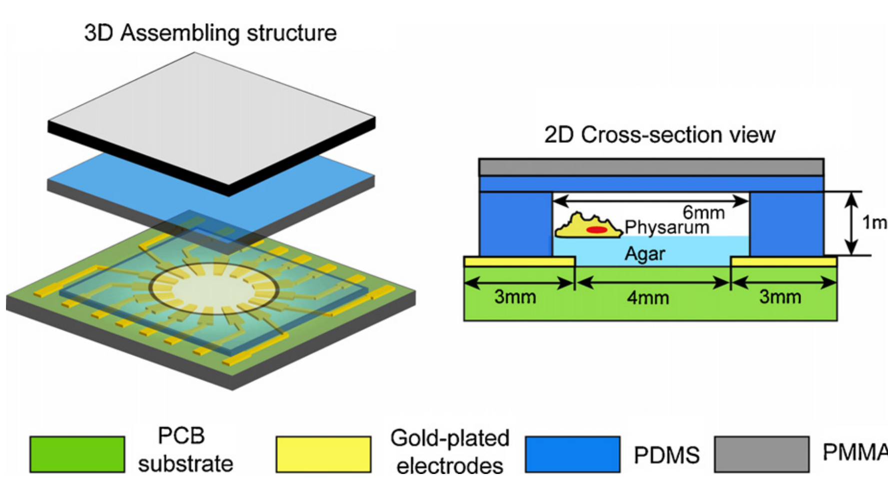


Fig. 2. Diagram showing the structure and cross-section of the EIT chip used to image *Physarum* in culture.

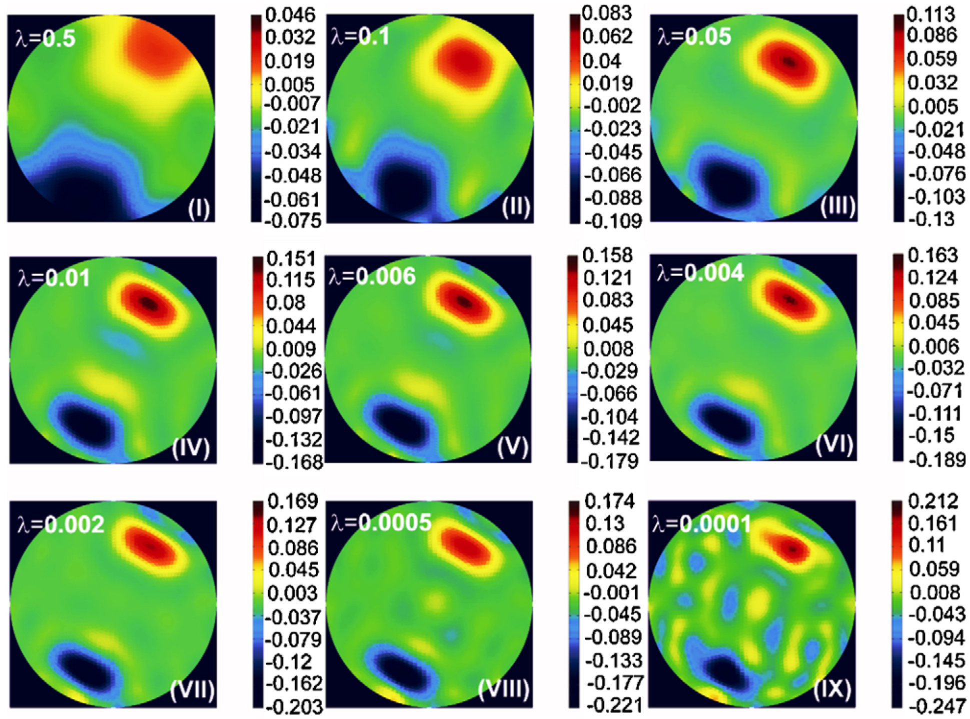


Fig. 3. Reconstructed images for two ellipsoidal phantoms for different values of *hyper-parameter*. Colour bars show the values of the reconstructed differential conductivity (S/m). Large *hyper-parameter* leads to coarse reconstructed images while too small *hyper-parameter* generates noise in the reconstructed images.

3.5. Image reconstruction

Images were reconstructed using EIDORS software, developed by Lionheart's group (Polydorides and Lionheart, 2002; Adler and Lionheart, 2006). The software consists of four primary objects: forward model, inverse model, data and image. The forward model uses Netgen as a mesh generator and defines the electrode positions, contact impedance, stimulation patterns, and boundary conditions. It also calculates the Jacobian matrix. In EIDORS, finite element method is used to solve the forward model, in which the domain of the system is discretized into small elements (e.g. triangular elements). The electrodes are represented as one segment of a triangular element along the boundary with two-point denoting the edges. However, the one dimensional segment model of the electrodes in EIDORS does not exactly represent the planar electrodes in the EIT chip. This leads to inaccuracy in modelling the

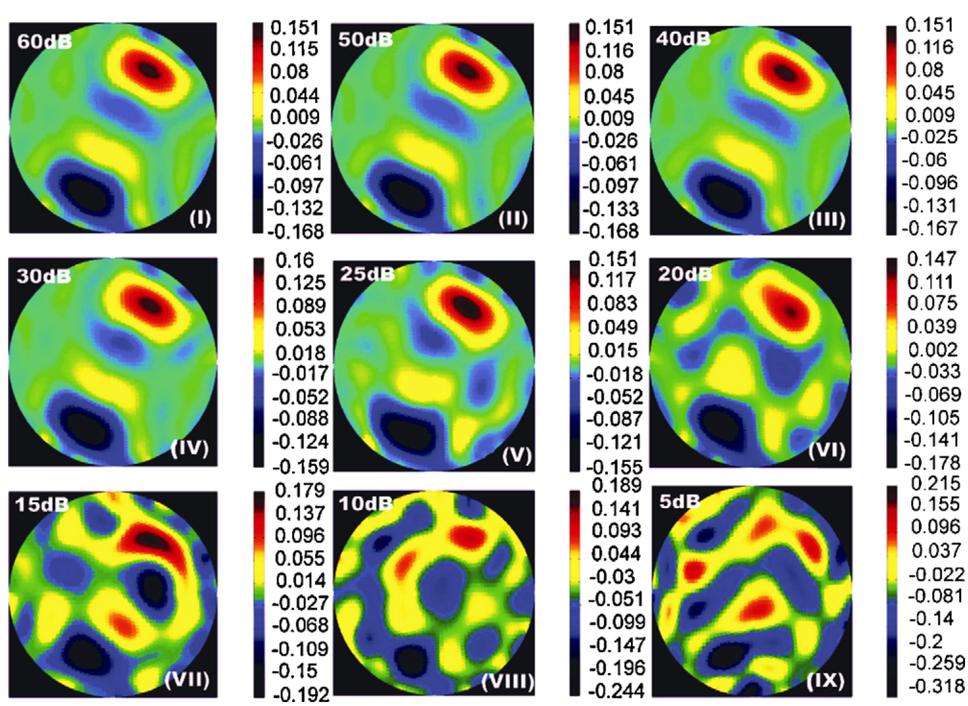


Fig. 4. Reconstructed images for two ellipsoidal phantoms with different levels of white noise. Colour bars show the values of the reconstructed differential conductivity (S/m). When the SNR of the system is less than 20 dB, the distortion, the ghost images and the noise representation in the reconstructed images can be observed.

position and dimension of the electrodes, which will generate distortion and noise in the reconstructed images. The inverse model has options for static image reconstruction (using only a single data object) and differential image reconstruction (using two data objects). The regularized image reconstruction algorithms require an *image prior*, which corresponds to the term R^*R in Eq. (4) and a choice of *hyper-parameter*, which corresponds to the parameter λ in Eq. (4). The data object contains the sets of simulated or measured voltage data. The image object displays the simulated or reconstructed conductivity values. In our work, the differential reconstruction scheme was used to minimize the influence of common sources (e.g. amplitude of the stimulating signal) on the reconstructed images. Therefore, two sets of data were simulated or measured for image reconstruction. One set is from the homogenous system (no perturbation in the conductivity) and the other set from the inhomogeneous system, containing the phantoms or cell.

4. Results and discussions

4.1. Simulated results

A finite element model of the EIT chip was set up in Comsol Multiphysics 3.4 (Comsol, Sweden). Two ellipsoidal phantoms (see supplemental material) were introduced into the background medium (conductivity 1 S/m). One inclusion has a higher conductivity (1.15 S/m) than the agar/medium and the other a lower conductivity (0.85 S/m). Refined mesh elements were used along the electrodes and the boundaries of the ellipsoids to improve the accuracy of the numerical simulations. To examine the fidelity of image reconstruction using EIDORS, the reconstruction results were analyzed and discussed in terms of the choice of hyper-parameter and noise interference.

The *hyper-parameter* λ, is a regularization factor in the reconstruction algorithm and is kept constant in each iteration. Adler and Guardo (1996) proposed a 'noise figure strategy' to determine the value of the *hyper-parameter*. In EIDORS, a default value of the *hyper-parameter* is set automatically from the forward model if not specified by the user. For the 2D circular 16-electrodes system, the default value of λ is 0.03. The choice of *hyper-parameter* has a significant impact on the reconstructed images as well as on the reconstructed conductivity. Fig. 3 shows reconstructed images for different values of *hyper-parameter*. The values of hyper-parameter are varied from 0.5 to 0.0001. Reconstructions were performed for 5184 mesh elements. It is observed that a smaller value of *hyper-parameter* increases the sensitivity to conductivity perturbations. However, if the *hyper-parameter* is too small, noise is generated in the reconstructed images.

White Gaussian noise was added to the simulated voltage data to evaluate the effect of noise on the reconstructed images. The reconstructed images (Fig. 4) were simulated for different signal-to-noise ratios (SNR), from 60 dB to 10 dB and the value of *hyper-parameter* is set to 0.01. Distortion in the reconstructed images is observed as the SNR decreases to 20 dB (noise level at 10% of the difference between the homogeneous data and inhomogeneous data). It also can be observed that EIDORS generates 'ghost images', mirroring the phantoms. A low conductivity ghost image is generated next to the reconstructed high conductivity phantom and *vice versa.* This effect (clearly shown in the colour figure) becomes significant if the SNR is low (SNR < 15 dB).

4.2. Experimental results

4.2.1. Impedance measurements of Physarum polycephalum

Little is known about the electrical properties of *Physarum*. Halvorsrud et al. (1995) monitored the rhythmic changes in *Physarum* from time-dependent resistance changes. To determine the conductivity of the organisms, the impedance of pure agar, agar plus *P. polycephalum*, and pure *P. polycephalum* was measured using a two-electrode configuration. The amplitude of stimulating signal was 100 mV and the impedance spectra (magnitude and phase) measured between 1 kHz and 1 MHz. The equivalent circuit model of this system is shown in Fig. 5a. The impedance of the electrical double layer (EDL) (Russel et al., 1989) at the interface between the electrode and the sample, Z_{EDL}, is modelled using a constant phase element (CPE):

$$Z_{EDL} = \left[T(i\omega)^p\right]^{-1} 0 \le p \le 1 \quad (5)$$

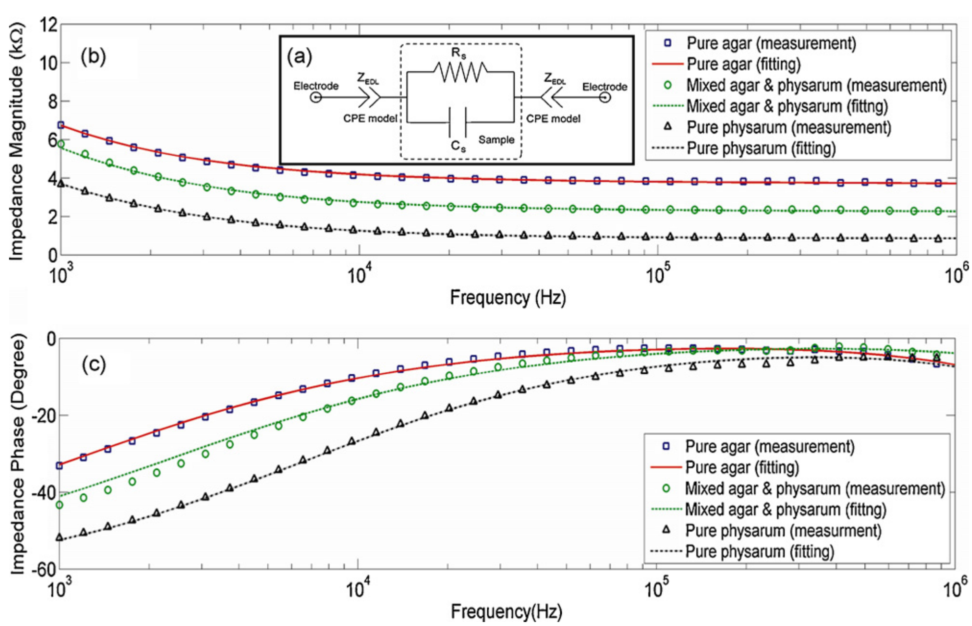


Fig. 5. (a) Schematic showing the equivalent circuit model of the two-electrode impedance measurements for evaluating the electrical properties of *Physarum polycephalum* and agar gel. (b) and (c) Plots showing the impedance spectra (measured and fitted) for three samples: pure agar, mixed agar and *Physarum*, pure *Physarum* between 1 kHz and 1 MHz. (b) Magnitude (c) phase. The achieved best fits in the three samples use the following parameters. For pure agar: $T = 4.96 \times 10^{-7}$, $p = 0.71$, $R_s = 3753\ \Omega$, $C_s = 4.8$ pF; for mixed agar and *Physarum*: $T = 4.69 \times 10^{-7}$, $p = 0.72$, $R_s = 2567\Omega$, $C_s = 4.5$ pF; for pure *Physarum*: $T = 4.02 \times 10^{-7}$, $p = 0.75$, $R_s = 877\ \Omega$, $C_s = 3.8$ pF.

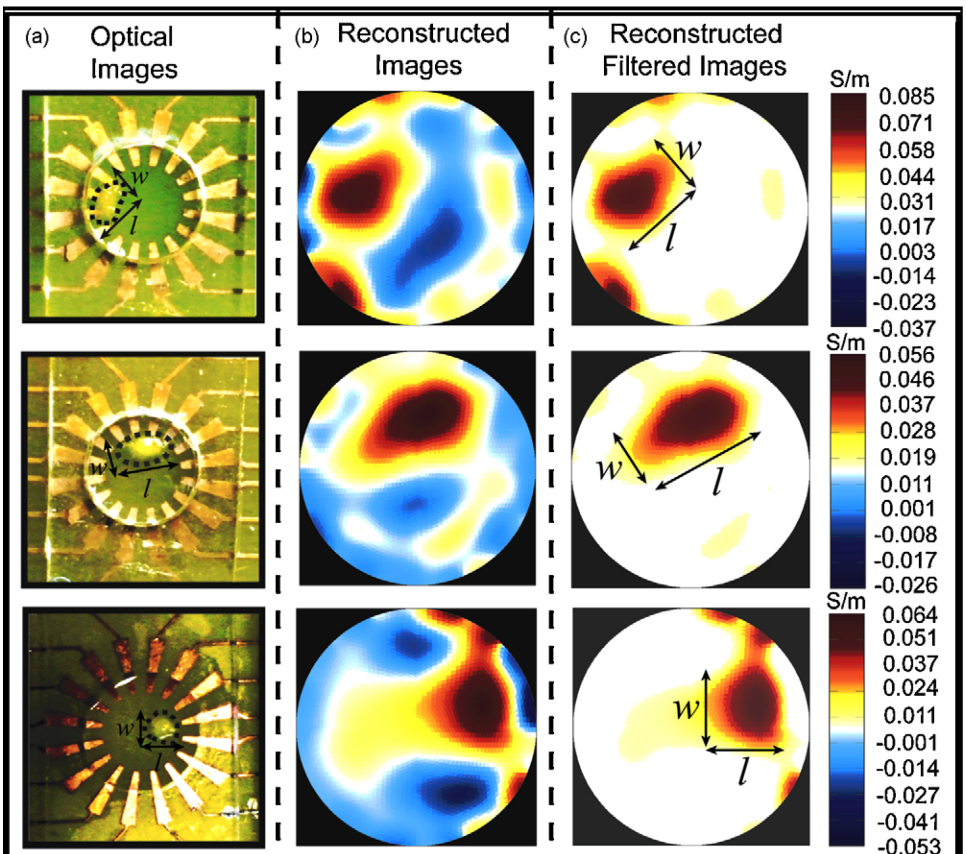
where $i^2 = -1$ and the constant T is a combination of properties related to both the surface and the sample, and independent of frequency. The exponent p is a measure of the electrode surface properties; when $p = 1$, the EDL is purely capacitive and is purely resistive when $p = 0$. The sample is modelled as a resistor R_s, and capacitor, C_s in parallel. The values of T, p, R_s and C_s were determined by fitting to the circuit model using ZView (Scribner Associates Inc.). As shown in Fig. 5b and c, the measured impedance spectra fit the equivalent circuit model for all the samples.

The conductivity of the agar gel (1.5% in DI water) was measured with a conductivity meter to be 17.7 mS/m. Fig. 5c shows that at 200 kHz the phase angle of the sample is close to zero degrees, indicating that at this frequency, the sample can be considered resistive. Therefore, this frequency was used for the *Physarum* cell imaging experiments. From the fitting parameters, the measured resistance of pure agar and *Physarum* are 3753 Ω and 877 Ω, respectively. From conductivity data for the pure agar, the geometrical "cell constant" of the two-electrode sample system is calculated as 66.4 m^{-1}, suggesting that the conductivity of *Physarum* is 75.8 mS/m, indicating that *Physarum* is more conductive than the agar. This means that the reconstructed image of *Physarum* should appear as regions of higher conductivity.

4.2.2. Conductivity imaging of Physarum polycephalum

Different shapes of *P. polycephalum* were grown on the surface of the agar in different positions within the chip as shown in Fig. 6a. Two sets of voltage data were measured using the adjacent method (stimulating signal = 100 mV @ 200 kHz). One set of data was taken for pure agar gel in a closed EIT chip, providing a homogeneous background. The second set of data was measured after *Physarum* was grown on the agar. Images were reconstructed using 5184 elements in the forward model. The initial value of *hyper-parameter* was set to 0.1 and adjusted in the range between 0.05 and 0.2 for smooth boundaries between *P. polycephalum* and the background.

Fig. 6b shows the corresponding reconstructed images from the voltage data. Brighter (red) corresponds to higher conductivity and darker (blue) to lower conductivity. The brighter region where the



	Top Physarum Sample			Middle Physarum Sample			Bottom Physarum Sample		
	Optical	Electrical	Relative Difference	Optical	Electrical	Relative Difference	Optical	Electrical	Relative Difference
Width (w)	1.41mm	1.45mm	2.8%	1.61mm	1.64mm	1.8%	1.76mm	1.82mm	3.4%
Length (l)	2.47mm	2.54mm	2.8%	3.19mm	3.27mm	2.5%	1.76mm	1.82mm	3.4%

Fig. 6. (a) Optical images of *Physarum polycephalum* on the agar gel in the EIT chips, (b) corresponding reconstructed images. The *Physarum* cell is more conductive than the agar gel, exhibiting as bright (red) region in the reconstructed images. (c) Filtered images, the ghost images are eliminated by filtering the low conductivity regions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of the article.)

Physarum cell locates (positive values in the reconstructed differential conductivity) can be clearly seen. The mirror ghost images are equivalent to regions of lower conductivity (negative values in the reconstructed differential conductivity), and produce variations in the reconstructed conductivity values. Because the ghost images are low conductivity regions, and the difference between *Physarum* and the background is positive, the ghost images were eliminated by setting a threshold in the reconstructed conductivity, as shown in Fig. 6c. There are other artefacts, exhibiting both high and low conductivity regions, distributed along the edges of the reconstructed images, which may come from the incompleteness of the one dimensional segment model of the electrodes in the forward model. This is similar to the reconstruction errors, reported by Linderholm et al. (2006), in which a point-electrode assumption was used.

To analyze the resolution of the reconstructed images, the dimensions (the width and the length) of the *Physarum* cell from the optical and electrical images of were compared. Dimensions of three samples (from top to bottom) are listed below the images in Fig. 6. The absolute difference in the dimensions between the optical and the reconstructed images ranges from 30 μm to 80 μm. The relative difference, calculated as the ratio of the absolute difference to the dimension in the optical image, shows that the minimum is 1.8% and the maximum is 3.4%. The average size of a single *Physarum* cell is approximately 2 mm in one dimension. Since the *Physarum* cells were imaged within a 6 mm diameter chamber, assuming linear scaling, it is anticipated that to image a single cell of 10 μm diameter, a 30 μm diameter chamber would be needed.

5. Conclusion

A miniaturized electrical impedance tomography system has been developed for non-invasive and non-contact imaging of cells with high temporal resolution. The chip consists of a circular array of 16-electrodes, and allows imaging of cells located at any position within the array. We have developed a finite element model to numerically analyze the effects of the regularization parameter in the reconstruction algorithm on the quality of the images, which is important in determining the fidelity of the reconstructed images. A reconstruction of the outline of *P. polycephalum* grown on agar has been obtained using low frequency measurements taken using the adjacent method. Our preliminary reconstructed images demonstrate the potential of developing a microscale electrical impedance tomography system for imaging single cells in culture from maps of the conductivity distribution in the system.

The developed system would be useful in diagnostic and bio-medical applications, such as cancer and cancer treatment monitoring for characterizing the spatial electrical properties of tumor cell colonies. Understanding the tumor growth is essential for screening programs, clinical trials and epidemiological investigations. Skourou et al. (2004) demonstrated the feasibility of using electrical impedance technique to detect small tumors (<3 mm) and tumor associated changes. The tumor growth varies but is generally from a few mm to tens of mm (Weedon-Fekjǽr et al., 2008), which falls into the imaging scale and resolution of our system. It can be expected that electrically mapping the electrical properties of tumour spatially will benefit the study in the dynamics in tumor cells spreading and proliferation.

Besides scaling-down the present system to micro level for a single cell imaging, our future work is also aimed at the optimization of the reconstruction algorithms and hardware instrumentation to improve the resolution of the image reconstruction.

Acknowledgements

We acknowledge Prof. Andy Adler for valuable discussions. We also thank to Jeff Gough and Paul Macey for building the multiplexing circuit board. T. Sun would like to acknowledge the postdoctoral fellowship from the National Centre for the Replacement, Refinement and Reduction of Animals in Research (NC3Rs), UK.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.bios.2009.09.036.

References

Adler, A., Guardo, R., 1996. IEEE Trans. Med. Imag. 15, 170–179.
Adler, A., Lionheart, W.R.B., 2006. Physiol. Meas. 27, S25–S42.
Borcea, L., 2002. Inverse Prob. 18, R99–R136.
Bayford, R.H., 2006. Annu. Rev. Biomed. Eng. 8, 63–91.
Bagshaw, A.P., Liston, A.D., Bayford, R.H., Tizzard, A., Gibson, A.P., Tidswell, A.T., Sparkes, M.K., Dehghani, H., Binnie, C.D., Holder, D.S., 2003. NeuroImage 20, 752–764.
Blasio, B.F.D., Laane, M., Walmann, T., Giaever, I., 2004. Biotechniques 36, 650–662.
Cherepenin, V., Karpov, A., Korjenevsky, A., Kornienko, V., Mazaletskaya, A., Mazourov, D., Meister, D., 2001. Physiol. Meas. 22, 9–18.
Chai, K.T.C., Hammond, P.A., Cumming, D.R.S., 2005. Sens. Actuators B 111-112, 305–309.
Chai, K.T.C., Davies, J.H., Cumming, D.R.S., 2007. Sens. Actuators B 127, 97–101.
Davalos, R., Runbinsky, B., Otten, D.M., 2002. IEEE Trans. Biomed. Eng. 49, 400–403.
Davalos, R., Otten, D.M., Mir, L.M., Runbinsky, B., 2004. IEEE Trans. Biomed Eng. 51, 761–767.
Daniel, J.W., Rusch, H.P., 1961. J. Gen. Microbiol. 25, 47–59.
Dove, W.F., Rusch, H.P., 1980. Growth and Differentiation in *Physarum polycephalum*. Princeton University Press, Princeton, NJ.
Fingerle, J., Gradmann, D., 1982. J. Membr. Biol. 68, 67–77.
Halvorsrud, R., Giaever, I., Laane, M.M., 1995. Protoplasma 188, 12–21.
Holder, D.S., 2005. Electrical impedance tomography: methods, history and applications. IOP Publishing, UK.
Hou, T.-C., Loh, K.J., Lynch, J.P., 2007. Nanotechnology 18, 315501.
Hato, M., 1979. Colloids Polym. Sci. 257, 397–405.
Keese, C.R., Wegener, J., Walker, S.R., Giaever, I., 2004. PNAS 101, 1554–1559.
Kaipio, J.P., Kolehmainen, V., Somersalo, E., Vauhkonen, M., 2000. Inverse Prob. 16, 1487–1522.
Kohn, R.V., Mckenney, A., 1990. Inverse Prob. 6, 389–414.
Linderholm, P., Braschler, T., Vannod, J., Barrandon, Y., Brouard, M., Renaud, Ph., 2006. Lab Chip 6, 1155–1162.
Linderholm, P., Vannod, J., Barrandon, Y., Renaud, Ph., 2007. Biosens. Bioelectron. 22, 789–796.
Linderholm, P., Marescot, L., Loke, M.H., Renaud, Ph., 2008. IEEE Trans. Biomed. Eng. 55, 138–146.
Mueller, J.L., Siltanen, S., Isaacson, D., 2002. IEEE Trans. Med. Imag. 21, 555–559.
Morgan, H., Sun, T., Holmes, D., Gawad, S., Green, N.G., 2007. J. Phys. D: Appl. Phys. 40, 61–70.
Pethig, R., 1979. Dielectric and Electronic Properties of Biological Materials. John Wiley & Sons Ltd., UK.
Polydorides, N., Lionheart, W.R.B., 2002. Meas. Sci. Technol. 13, 1871–1883.
Russel, W.B., Saville, D.A., Schowalter, W.R., 1989. Colloidal Dispersions. Cambridge University Press, Cambridge.
Schwan, H.P., 1957. Adv. Biol. Med. Phys. 5, 147–209.
Sun, T., Holmes, D., Gawad, S., Green, N.G., Morgan, H., 2007. Lab Chip 7, 1034–1040.
Santosa, F., Vogelius, M., 1990. SIAM J. Appl. Math. 50, 216–243.
Sauer, H.W., 1982. Developmental Biology of Physarum. Cambridge University Press, Cambridge, UK.
Sudbery, P.E., Grant, W.D., 1976. J. Cell Sci. 22, 59–65.
Skourou, C., Hoopes, P.J., Strawbridge, R.R., Paulsen, K., 2004. Physiol. Meas. 25, 335–346.
Tsuda, S., Zauner, K.-P., Gunji, Y.-P., 2007. Biosystems 87, 215–223.
Wegener, J., Keese, C.R., Giaever, I., 2000. Exp. Cell Res. 259, 158–166.
Wohlfarth-Bottermann, K.E., 1979. J. Exp. Biol. 81, 15–32.
Weedon-Fekjǽr, H., Lindqvist, B.H., Vatten, L.J., Aalen, O.O., Tretli, S., 2008. Breast Cancer Res. 10, R41.
Yorkey, T.J., Webster, J.G., Tompkins, W.J., 1987. IEEE Trans. Biomed. Eng. 11, 843–852.